Lab Task # 2

Note: Submission guidelines are provided at the end. Failing to follow the guidelines will result in marks deduction.

Scenario: Painting Hire Business

Problem Description:

A local businesswoman has decided to start her own Internet business, called Masterpieces Ltd, hiring paintings to private individuals and commercial companies. Because of your reputation as a database designer, she has called upon your services to design and implement a database to support her new business. At the initial planning meeting, to discuss the design, the following user requirements were requested.

The system must be able to manage the details of customers, paintings and those paintings currently on hire to customers. Customers are categorized as B (bronze), S (silver), G (gold) or P (platinum). These categories entitle a customer to a discount of 0%, 5%, 10% or 15% respectively.

Customers often request paintings by a particular artist or theme (e.g., animal, landscape, seascape, naval, still-life, etc). Over time a customer may hire the same painting more than once.

Each painting is allocated a customer monthly rental price defined by the owner. The owner of the painting is then paid 10% of that customer rental price. Any paintings that are not hired within six months are returned to the owner. However, after three months, an owner may resubmit a returned painting. Each painting can only have one artist associated with it.

Enter 10 dummy entries for each table at least to make appropriate queries.

1) Tables

Note: All IDs are Primary Keys in their respective tables.

a) Customers:

- CustomerID (AutoNumber)
- FirstName (Short Text)
- LastName (Short Text)
- Email (Short Text)
- Phone (Short Text)
- Address (Short Text)
- Category (Short Text)
- Discount (Number)

b) Paintings:

- PaintingID (AutoNumber, PK)
- Title (Short Text)
- ArtistID (Number, FK)
- Theme (Short Text)
- MonthlyRentalPrice (Currency)
- OwnerID (Number, FK)
- Status (Short Text, Lookup: 'Available', 'Hired', 'Returned')
- LastHiredDate (Date/Time)
- Returned (Lookup, Yes/No)
- ReturnDate (Date/Time) {NULL if not returned}

c) Artists:

- ArtistID (AutoNumber, PK)
- FirstName (Short Text)
- LastName (Short Text)
- Bio (Long Text)

d) Owners:

- OwnerID (AutoNumber, PK)
- FirstName (Short Text)
- LastName (Short Text)
- Email (Short Text, Indexed: Yes, No Duplicates)
- Phone (Short Text)

e) HireTransactions:

{Tracks painting hire records, including current and past rentals}

- TransactionID (AutoNumber, PK)
- CustomerID (Number, FK)
- PaintingID (Number, FK)
- HireStartDate (Date/Time)
- HireEndDate (Date/Time, Allow Nulls for ongoing rentals)
- TotalCost (Currency)

2) Relationships

Define relationships with referential integrity:

- **Customers ↔ HireTransactions:** A customer can hire multiple paintings.
- **Paintings ↔ HireTransactions:** A painting can appear in multiple hire transactions.

- **Paintings ↔ Artists:** An artist can create multiple paintings.
- **Paintings ↔ Owners:** An owner can own multiple paintings.

3) Forms:

Create Forms for tables: Paintings, Customers, Artists, HireTransactions.

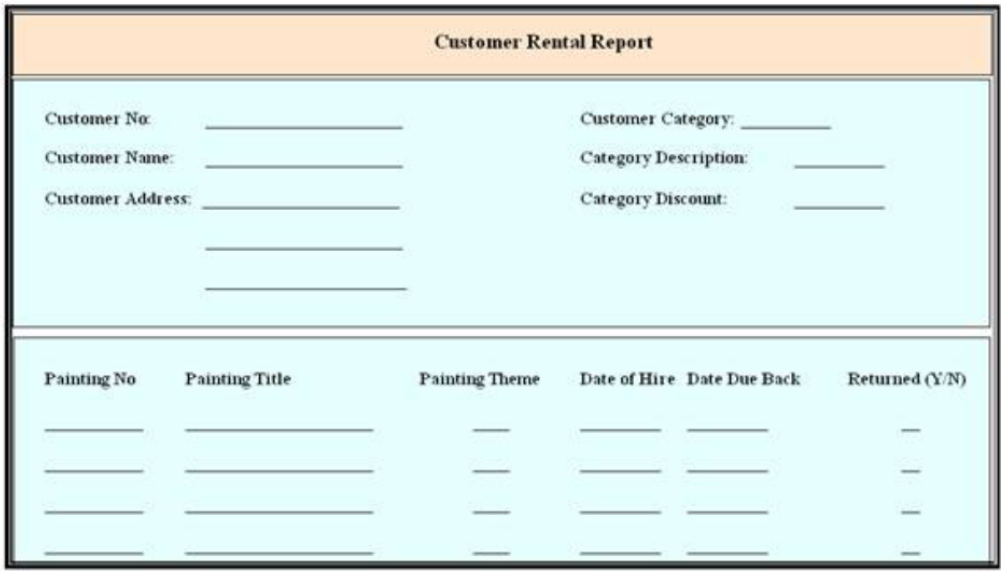
4) Queries

I. Count the total number of paintings in each theme.
II. List all hire transactions where paintings are currently not returned.
III. Find all customers with "Gold" membership who hired more than 5 paintings.
IV. Use BETWEEN to filter hires made in a specific date range.
V. Calculate the total revenue generated by customers in each membership category.

4) Reports

Several reports are required from the system. Three main ones are:
1. For each customer, a report showing an overview of all the paintings they have hired or are currently hiring
2. For each artist, a report of all paintings submitted for hire
3. For each artist, a returns report for those paintings not hired over the past six months.

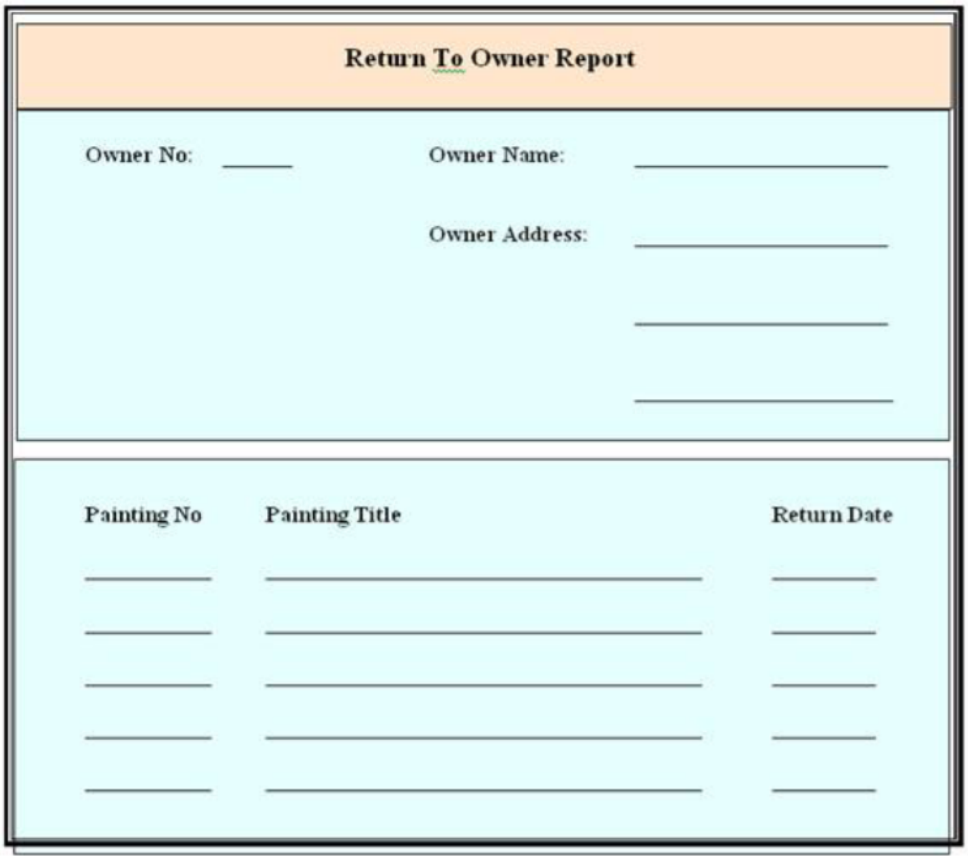
Customer Rental Report



Artist Report



Return to Owner Report



NOTE: The reports should be visually appealing.

Submission Instructions:

1. First, rename your file as i23xxxx_FullName_Section.accdb.
2. Prepare a PDF file with all the screenshots and proper headings.
3. Submit both the PDF and database file on GCR (Don't submit ZIP file)
3. Please ensure that your submission is on time; late submissions will **not be accepted**.